Trace Tables

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Transparent Book Recommender

Trace Tables

> Prepare Trace Tables: check the target, prerequisites, resources and adaptable teaching schedule.

Teach the learning cycle

Primary target: **Trace variable** values before and after each **loop iteration**

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min

> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Case exposure check: this source example appears in ai-2/4, ai-2/5, python-bridge/6. Reuse a familiar case for recall or modelling; verify that the case used to assess transfer has not already received feedback.

Retrieval answer guidance

Branching Logic

Without opening the old lesson, explain one key idea from Branching Logic. Give an example and a mistake that the idea helps you avoid.

Look for: **Trace** the selected branch in an if elif else chain

Reference example, not the only acceptable wording: if tests negative first; elif checks <= 12 only after that fails; else handles the rest.

Unindented print runs after any selected branch, whose assignment gives label a value.

Precision in Algorithms

Without opening the old lesson, explain one key idea from Precision in Algorithms. Give an example and a mistake that the idea helps you avoid.

Look for: Translate a short paper calculation into a Python script

Reference example, not the only acceptable wording: packs = 4; per_pack = 6; total = packs * per_pack; print(total), each on its own line. The first two bind quantities, the third computes 24 and the fourth displays 24.

Assignment alone produces no displayed **output** in this script.

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

Start score=1; add 4; if score>3 subtract 2; double score. Record score after every operation. A student evaluates the condition using the initial value. Locate the divergence.

Concept to explain

A **trace table** records changing **state** and reveals reasoning errors.

Term	Meaning
variable	A named reference to a value that can be used or changed in a program.
state	The current values and information that determine what a process does next.
trace	A record of the successive states, decisions or information transformations in a process.
iteration	One repeat of a process, often followed by inspection and revision.

Prepare the class

Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.

Alternative compact allocation

> This compact option is not the full-week schedule. Use the main teaching cycle and allow extra sessions for the investigation, independent work and project.

Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

Teach for **loop**s with finite integer ranges before adding more **loop** forms. Keep range values, body executions and **output** events separate. Reference: https://docs.python.org/3/tutorial/controlflow.html#the-range-function.

The examples use a positive default step and do not require lists or while **loop**s.

Readiness answer

It reads the old 2, adds 3 and stores 5. Assignment updates a value; it is not an algebraic equality.

Python 3 editor and **loop**s.py

Counters or numbered paper cards

Blank **trace table** with turn total before and total after columns

End goal for the pictorial case

By the end, I can make a **trace table** for a for-**loop**, including zero **iteration**s.

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

Explain the mechanism

> Model Trace Tables using the supplied case, labelled mechanism and reasoning. Make the scope of each conclusion explicit.

Input → Process → Output

Trace the information. At each stage, ask what changed and how you could test it.

Which stage could fail even when the other two are correct? Explain:

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.

2. Correct answer: score = score + 3 → 5.

3. Hint used: score = score – 1 → 4.

4. A table shows exactly where an unexpected value first appears.

Recalculate, re**trace**, or re**state** the reasoning in your own representation:

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

Concrete teaching case

The teaching case

Set score=2; add 3; if score>4 subtract 1; double score.

The teaching case

Trace every value.

Exact explanation to model

> **Why the result follows**
>
> 2,5,4,8. The comparison is evaluated at 5 before subtraction.

Explain the added worked example

Explain the added worked example · 1

Imagine adding numbered counters: first 1, then 2, then 3. A **loop** can repeat the same addition with a different number each time. This week you will predict each change before running a short Python script.

Tracing repeated calculations prepares you to inspect the data **process**ing used in larger AI applications.

Explain the added worked example · 2

In for turn in range(1, 4), Python supplies 1, 2 and 3 in order. The start is included and the stop is excluded. The indented body runs once for each supplied value. The colon and four-space indentation mark that body.

This lesson uses integer range values with the default positive step of 1.

The **output** is 6. Before the first **iteration**, total is 0. Each assignment reads the previous total and adds the current turn. After turn 3, the range is exhausted, so execution continues after the **loop**.

The table records **state**; it does not create extra instructions.

Key terms

Term	Meaning
Loop	A block of instructions that can run repeatedly
Iteration	One execution of a loop body
Accumulator	A variable that keeps a running result
Initialization	Giving a variable its starting value

Term	Meaning
Trace table	A record of variable values as instructions execute

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Range expression	Values supplied	Number of iterations
range(1, 4)	1, 2, 3	3
range(1, 2)	1	1
range(1, 1)	No values	0
range(4)	0, 1, 2, 3	4

Model the reasoning aloud · 3

With one argument, range starts at zero. range(4) supplies four values but none is 4. If the worked script uses range(1, 1), the body never runs and the initialized total remains 0.

Do not rely on a **variable** first assigned inside a **loop**: a zero-**iteration** run might never assign it.

> **Replace range(1, 4) with range(1, 3) in the worked script. Predict the supplied values and final total. What stop value is needed to include 3?**

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Trace Tables

A fictional reading-token jar starts with one token and gains three per round.

The facts we will use

> **Supplied facts**
>
> Python: total=1; for turn in range(4): total=total+3; after the **loop** print(total).

> **Supplied facts**
>
> The update is inside the **loop**; print is outside.

AI ACADEMY / AI-2/4

Loop totals after each iteration

	turn	total after update
First	0	4
Second	1	7
Third	2	10
Fourth	3	13

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-2/4

Trace Tables

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Initial state

Before the loop, total=1.



2. Iteration values

range(4) supplies turn values 0,1,2,3.



3. Updated totals

After each turn total is 4,7,10,13.



4. After the loop

Print 13 once, after all four updates.

CHANGE ONE THING

Use range(0) instead of range(4).

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Initial state	Before the loop, total=1.
2. Iteration values	range(4) supplies turn values 0,1,2,3.
3. Updated totals	After each turn total is 4,7,10,13.
4. After the loop	Print 13 once, after all four updates.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

The turn value labels an **iteration**; it is not the number added. The fixed update adds three each time. Four **iteration**s take the total from 1 to 13.

Keeping print outside the **loop** produces one final display rather than four intermediate displays. Initialise total before the **loop** so a zero-**iteration** run still has a defined value.

What this example does not establish

Keep the stated limit

This finite **loop** does not establish behavior of a while-**loop** with a different update or

stopping condition.

Change one thing in the pictured case

Change one thing

Change: Use range(0) instead of range(4).

Predicted result

Prediction: The **loop** body never runs and the printed total is 1.

Why the result follows

Reason: The iterable contains no turn values; the initial assignment still happened.

Ask while showing the picture

Ask while showing the picture · 4

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

TEACHER GUIDE · 3 OF 9

Demonstrate and guide practice

> Demonstrate Trace Tables, reduce support deliberately and check individual reasoning before discussing answers.

Board sequence

1. Put the facts on the board

Write the given facts: Set score=2; add 3; if score>4 subtract 1; double score.

2. State the decision to explain

Write the question to explain: **Trace** every value.

3. Reveal the worked reasoning

Reveal after **prediction**s: 2,5,4,8. The comparison is evaluated at 5 before subtraction.

Hold back the independent task

Keep for an independent attempt: Set score to 2. Add 3, then double the result. Record the score after each operation. What changes if you double before adding?

Classroom dialogue

Ask the class

Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.

Language to model

Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, **trace** conditions for decisions, and identify supporting sources for claims.

In pairs, one learner solves while the other checks **evidence**; exchange roles. Remove the worked answer before independent assessment.

Model these guided questions

range(1, 3) supplies 1 and 2, giving total 3. Stop 4 includes 3. Exit: the reset bug agrees for one **iteration** because there are no earlier additions to lose. range(1, 4) should give 6; the faulty version gives 3.

Case-specific teaching sequence

Stage	Teacher action
Session 1 start to 7 minutes	Retrieve assignment using the prerequisite.
7 to 20 minutes	Model range with numbered cards and complete W1.
20 to 33 minutes	Trace each row of the worked addition and predict output.
33 to 45 minutes	Complete W2 and compare predicted and observed values.
Session 2 start to 8 minutes	Retrieve included start and excluded stop.
8 to 20 minutes	Trace the reset bug, complete W3 and discuss the guided response.
20 to 35 minutes	Collect independent W4 code and zero one and several iteration traces.
35 to 45 minutes	Use W5 and the exit to examine assumptions and revealing tests.

Reduce help deliberately

Completion practice

Completion task: Copy the worked script. Change the range to range(2, 5). Record turn, total before and total after each **iteration**. Predict the single **output**, then run and compare.

Reasoning to listen for

Teacher check: Rows are 2 / 0 / 2, 3 / 2 / 5, 4 / 5 / 9. Final **output** 9. Keep the observed result distinct from the **prediction**.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full **trace** only when the partial **trace** is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual reasoning check

Use the worked case for **Loop**s and **trace table**s. Proposed explanation: “The stop value is included”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Expected repair

Expected correction: List the supplied values before tracing; range(1, 4) excludes 4.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictorial practice question

Move print inside the original **loop**. What sequence is displayed?

Reasoned teacher answer

4,7,10,13 on separate lines, because each print follows one update.

Changed-case reasoning: The **loop** body never runs and the printed total is 1. The iterable contains no turn values; the initial assignment still happened.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Run the weekly evidence lab

> Run the Trace Tables investigation and use the first unsupported step to choose a teaching response.

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.

Use columns for step, event, old score, new score, and branch. **Trace** once by hand, predict the next row, then compare.

> What did the hardest test teach you that the normal test did not?

Make the lab runnable

Use the concrete case as the shared normal case and the independent question as a second case. Each pair creates a boundary or missing-information case and predicts its result before testing.

Keep the expected result separate from an observed mistake.

Record	What to capture
Input or evidence	Exact data, instruction or claim
Expected result	Answer or safe fallback declared before the run
Observed result	Actual result, including errors
Explanation	Rule, calculation or evidence explaining the outcome
Next test	One justified change and a new prediction

Diagnose and reteach

For guessing, return to given facts and request a **trace**. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation.

Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

The stop value is included: List the supplied values before tracing; range(1, 4) excludes 4.

Initialization belongs inside every **iteration**: Resetting the **accumulator** loses earlier work.

An unindented print repeats with the body: Indentation determines which **state**ments repeat.

Zero **iteration**s mean the entire program does nothing: **Initialization** and **state**ments after the **loop** still execute.

Match the support to the first error

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.

What the response shows	Next teaching action
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

Assess and explain the answers

> Use the Trace Tables questions, teacher keys and criteria after the relevant attempts. Previously discussed cases are practice; protect the separately selected independent task.

Worked case answer

Teacher key · Worked case answer

2,5,4,8. The comparison is evaluated at 5 before subtraction.

Independent question

Independent question · withhold the key

Set score to 2. Add 3, then double the result. Record the score after each operation. What changes if you double before adding?

Expected independent answer

> **Teacher key · Expected independent answer**
>
> The **trace** is 2, 5, 10. Doubling first gives 2, 4, 7. Operation order changes the result.

What counts as evidence

Criterion	0 points	1 point	2 points
Result	Incorrect or absent	Partly correct	Correct with necessary conditions
Reasoning	No supporting trace	Some steps	Clear mechanism, calculation or source support
Limit	Overclaims certainty	Vague limitation	Relevant uncertainty or missing fact
Transfer	Copies without adapting	Adapts with prompts	Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold.

Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the **evidence** that challenges this claim: “A **trace table** records only the final answer.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

> **Teacher key · Original transfer answer**
>
> **Trace** is 1,5,3,6. The condition sees 5, so subtraction runs. Using the initial 1 incorrectly skips subtraction and gives 10. A **variable**'s current **state**, not its first value, determines each operation.

Explanations for the additional workbook examples

W1 Explain the range

Workbook question · W1

List the values supplied by range(2, 5) and explain why 5 is absent. How many times does its body run?

Teacher reasoning · W1

2, 3 and 4; the stop value is excluded. The body runs three times.

W2 Trace before running

Workbook question · W2

Copy the worked script. Change the range to range(2, 5). Record turn, total before and total after each **iteration**. Predict the single **output**, then run and compare.

Teacher reasoning · W2

Rows are 2 / 0 / 2, 3 / 2 / 5, 4 / 5 / 9. Final **output** 9. Keep the observed result distinct from the **prediction**.

W3 Repair the reset

Workbook question · W3

For the faulty script with range(1, 4), record total after each addition. Explain the wrong final **output** and remove only the faulty reset.

Teacher reasoning · W3

Faulty after-addition values are 1, 2, 3 because each starts from zero. Remove the indented total = 0. Correct after-addition values are 1, 3, 6; final **output** 6.

W4 Build a repeated addition independently

Workbook question · W4

A box starts with 5 counters. Each round adds 2 counters. Use rounds = 3, an initialized total and for turn in range(rounds). Print the final total once after the **loop**.

Trace the totals and predict fresh runs with rounds 0, 1 and 3. Assume rounds is a nonnegative integer.

Teacher reasoning · W4

Initialize total = 5 before the **loop**; add 2 inside it. For three rounds, turn values 0, 1, 2 give totals 7, 9, 11. Final **output**s for 0, 1, 3 rounds are 5, 7, 11. Print is unindented.

W5 Check a contract limitation

Workbook question · W5

In your counter script, range(-2) performs no **iteration**s. Does returning the initial 5 make negative rounds valid? Explain what the script checks and what it assumes.

Teacher reasoning · W5

No. The written contract requires a nonnegative integer. The script assumes this and does not reject negative values. Empty range behavior does not validate the **input**; report the limitation or add an explicit check.

Assess independent transfer

Independent question · withhold the key

Selected case: A box starts with 5 counters. Each round adds 2 counters. Use rounds = 3, an initialized total and for turn in range(rounds). Print the final total once after the **loop**.

Trace the totals and predict fresh runs with rounds 0, 1 and 3. Assume rounds is a nonnegative integer.

Selected case answer

Teacher key · Selected case answer

Initialize total = 5 before the **loop**; add 2 inside it. For three rounds, turn values 0, 1, 2 give totals 7, 9, 11. Final **output**s for 0, 1, 3 rounds are 5, 7, 11. Print is unindented.

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Teacher key · Selected case answer

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible **trace**.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

Connect and extend learning

> Connect Trace Tables to Transparent Book Recommender, adapt support and plan a later changed-case check.

Transparent Book Recommender

Trace each book's genre, length, difficulty, and total score before sorting.

Keep the decision, **input**, result, explanation and remaining limitation in the project **evidence** record. Retrieve this record next week rather than restarting the project.

Support and stretch

Trace one record or condition at a time with counters and labelled columns. Stretch: introduce a tie, boundary or missing field and declare its handling policy before rerunning.

Exit reflection

Ask for one decision the learner can now justify and one situation requiring more **evidence** or human review. Confidence ratings are separate from correct independent performance.

Bridge to the next week

Next: Week 5: **Loop**s and Stopping. Retrieve today's **evidence** and identify what carries forward and what is new.

Support and fresh reassessment

Use counters and let the learner move a pointer through one instruction at a time. Offer a blank **trace** grid and permit oral explanation.

Extension: put print inside the correct **loop** and explain why zero rounds produces no **output** while an outside print still does.

W4 must preserve the initial 5, update by 2, **trace** three **iteration**s and print once. Require **output**s 5, 7, 11 for zero one and three rounds.

After support, retry with an initial 4 and additions of 3 for zero one and two rounds: **output**s 4, 7, 10. Record assistance before judging independence.

Week 7 stores several values in a list and distinguishes an index from the value at that position.

Plan the delayed return

Suggested return points in this level: ai-2/5, ai-2/7, ai-2/10. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Project evidence from this case

Trace recommender **loop**s with current item, eligibility, score and current best so each update can be explained.

TEACHER GUIDE · 7 OF 9

Prepare, teach and assess

> Use this preparation summary alongside the detailed Trace Tables lesson; it is a checklist rather than a replacement for the full sequence.

Teaching target

Teaching target: **Trace variable** values before and after each **loop iteration**

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. No paid AI account is required for the paper reasoning activities.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min

> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

Before class, solve the supported practice: Copy the worked script. Change the range to range(2, 5). Record turn, total before and total after each **iteration**. Predict the single **output**, then run and compare.

Supported practice answer

Reasoned practice answer: Rows are 2 / 0 / 2, 3 / 2 / 5, 4 / 5 / 9. Final **output** 9. Keep the observed result distinct from the **prediction**.

Check every learner before discussion: Use the worked case for **Loop**s and **trace table**s. Proposed explanation: “The stop value is included”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Repair the misconception

Misconception repair: List the supplied values before tracing; range(1, 4) excludes 4.

Support without giving away the answer

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the

reasoning goal.

Extend the reasoning

Extension: Use range(0) instead of range(4). Require a **prediction** and a reason before comparing. Expected result: The **loop** body never runs and the printed total is 1.

Reason: The iterable contains no turn values; the initial assignment still happened.

Protect the independent attempt

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint

Project checkpoint: Frame the need. Name the user, the problem, allowed **input**s and a safe simple starting method.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

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Use feedback, labs and repair paths

> Use feedback, earlier review and portfolios purposefully. Apply the coding routine only when this lesson has an executable lab.

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, **trace**/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

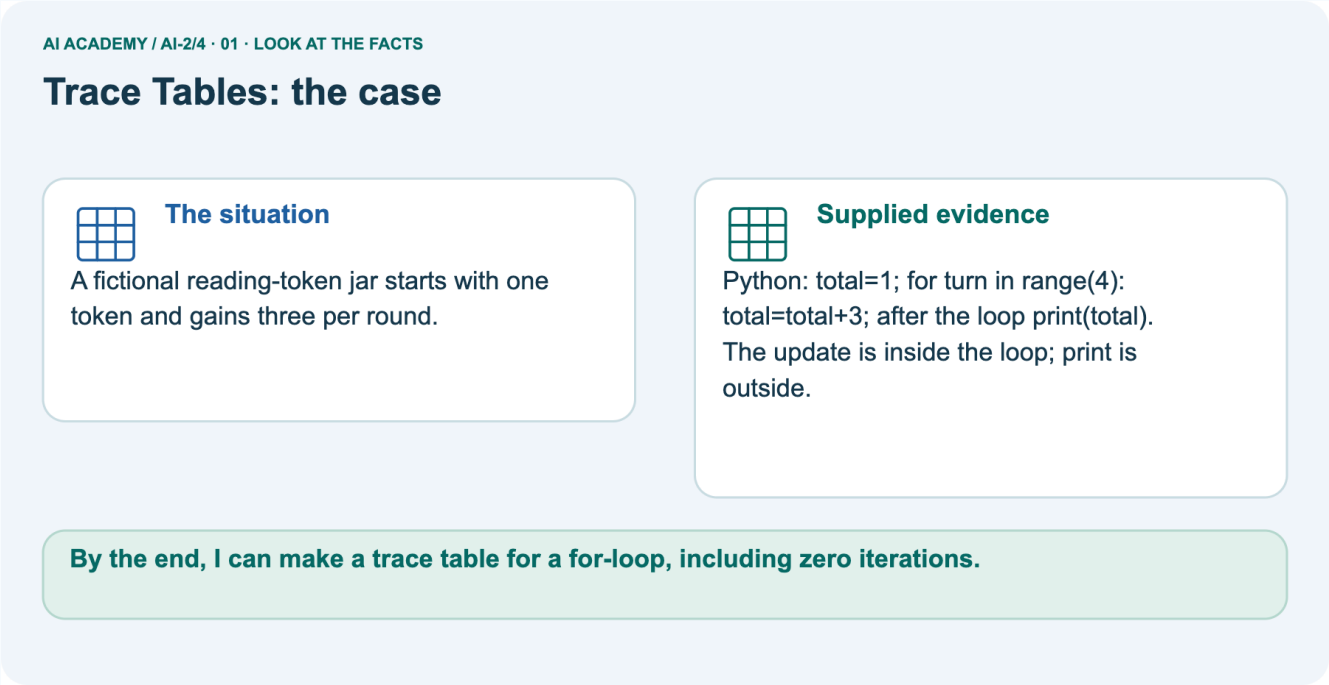
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Visual concept journey

> Show one Trace Tables graphic at a time, request a prediction and discuss its trace before the independent assessment.

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Trace Tables: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

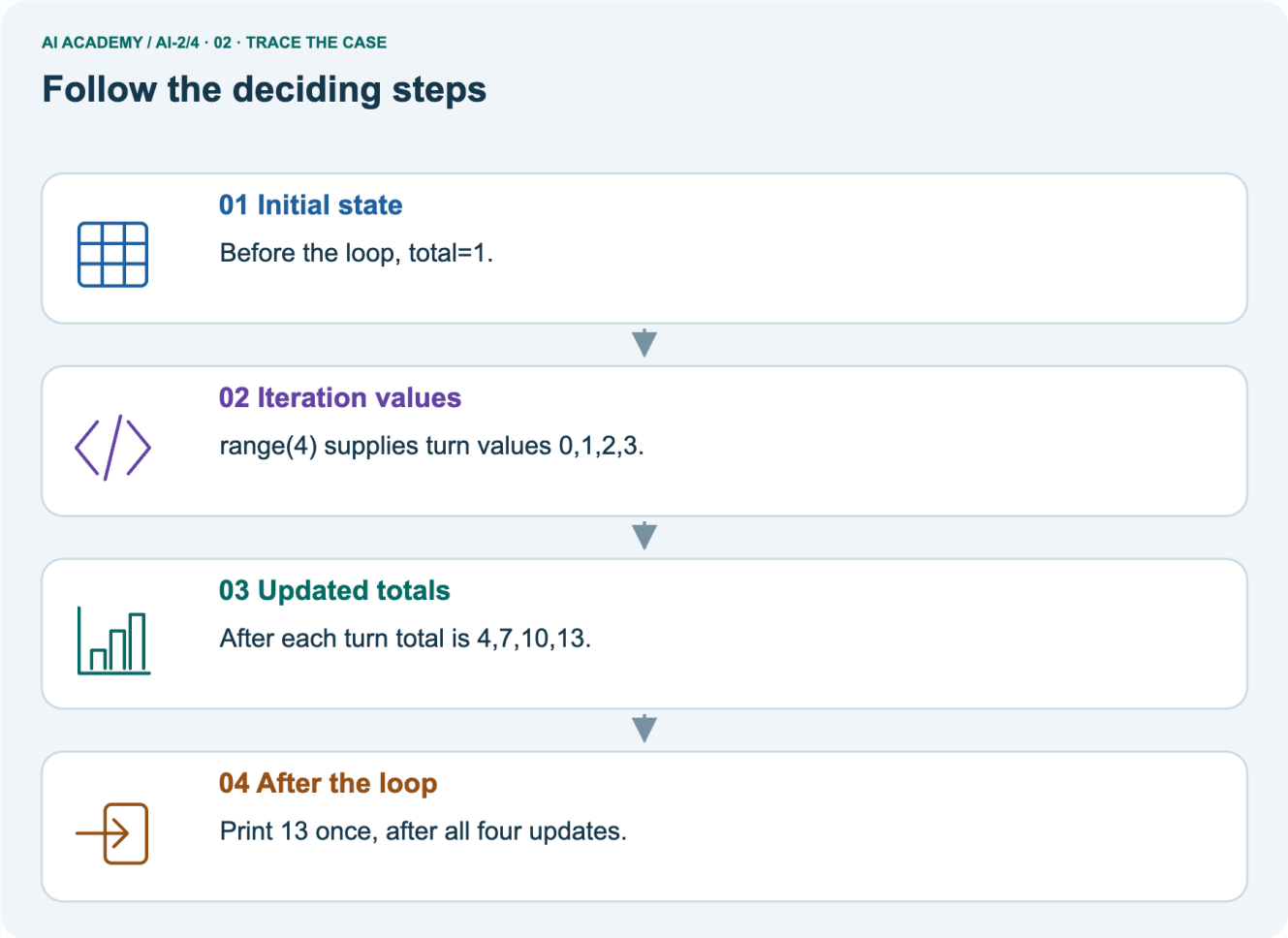
A fictional reading-token jar starts with one token and gains three per round. Supplied facts: Python: total=1; for turn in range(4): total=total+3; after the **loop** print(total). The update is inside the **loop**; print is outside.

> Point to the given fact that matters most. Explain why it matters.

Discuss while showing the picture

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Initial **state**: Before the **loop**, total=1. → **Iteration** values: range(4) supplies turn values 0,1,2,3. → Updated totals: After each turn total is 4,7,10,13. → After the **loop**: Print 13 once, after all four updates.

> Point to each numbered stage and explain which recorded **input** it uses.

Discuss while showing the picture

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Loop totals after each iteration



Follow the labelled picture with the matching explanation. Open the vector diagram

Loop totals after each **iteration**. First: turn = 0, total after update = 4. Second: turn = 1, total after update = 7. Third: turn = 2, total after update = 10. Fourth: turn = 3, total after update = 13.

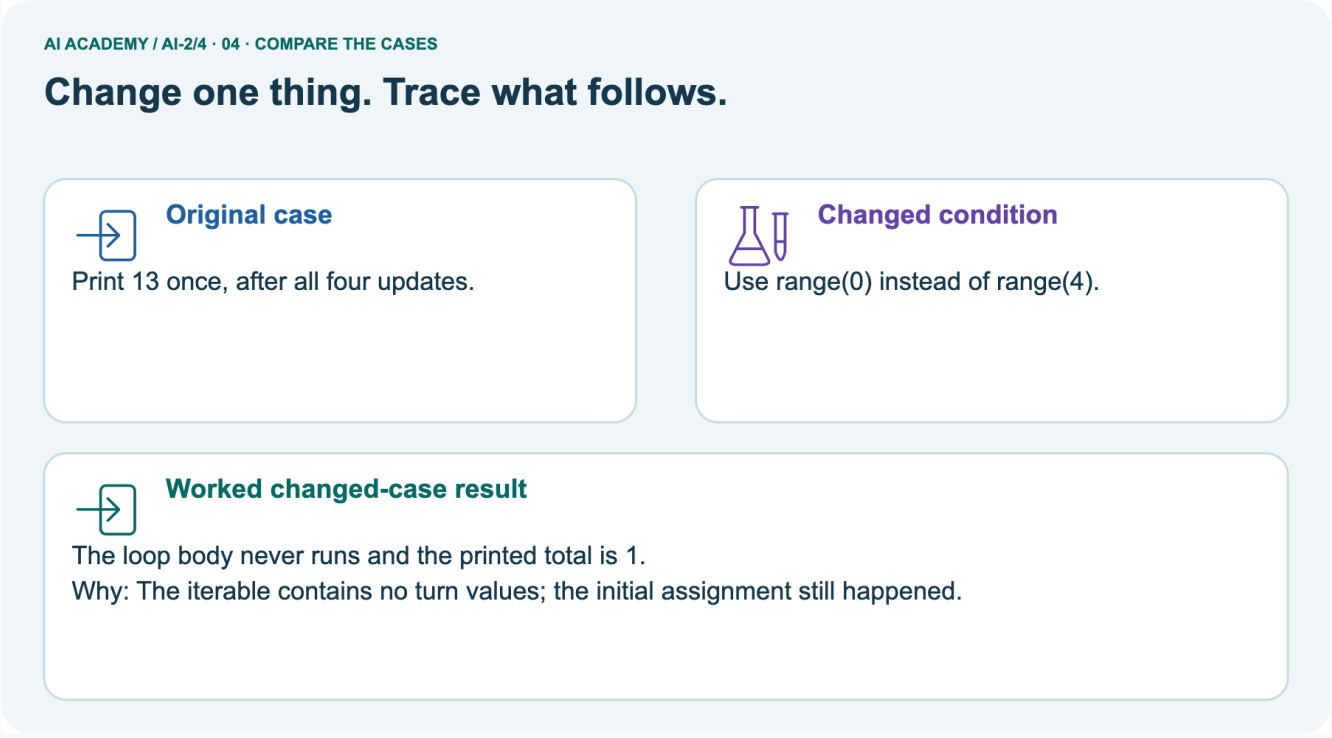
Read the labels and values. Explain the relationship the model shows.

Discuss while showing the picture

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Check row/column meanings, units and denominators before discussing a difference. Supplied toy values are not real class or model results.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Print 13 once, after all four updates. Changed condition: Use range(0) instead of range(4). Expected result: The **loop** body never runs and the printed total is 1.

Reason: The iterable contains no turn values; the initial assignment still happened.

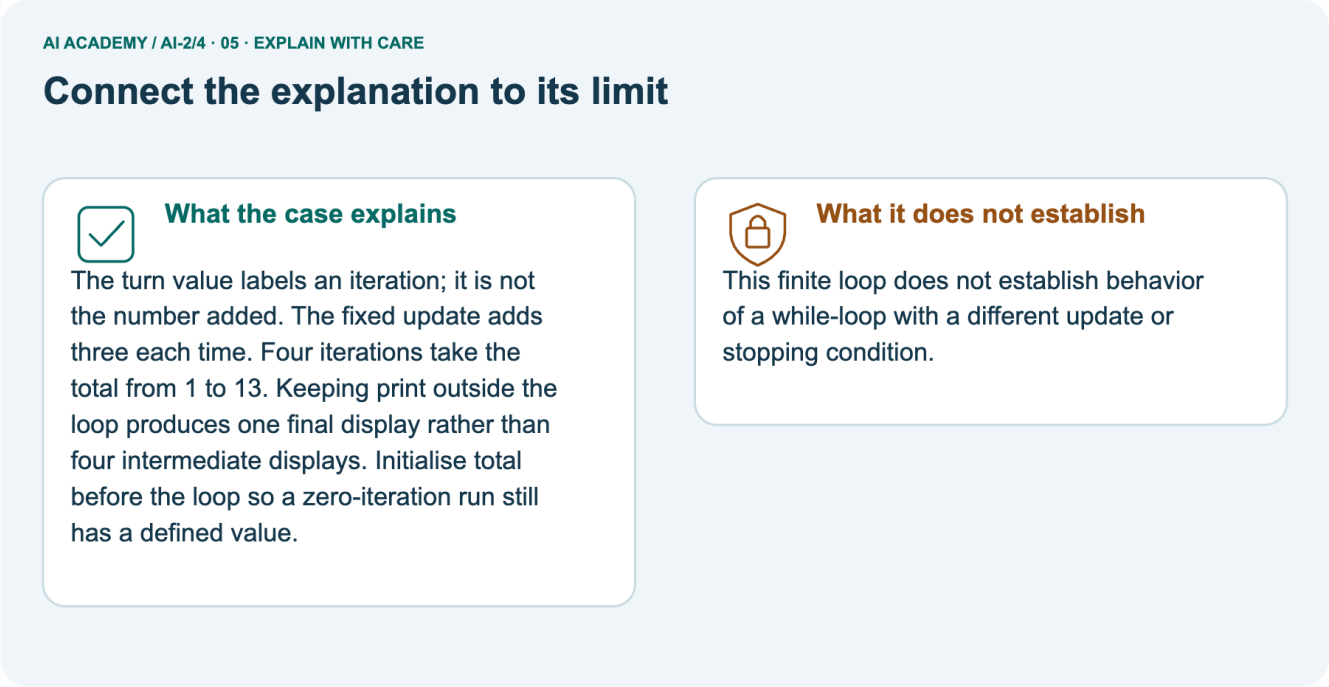
Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Discuss while showing the picture

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: The iterable contains no turn values; the initial assignment still happened.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

The turn value labels an **iteration**; it is not the number added. The fixed update adds three each time. Four **iteration**s take the total from 1 to 13.

Keeping print outside the **loop** produces one final display rather than four intermediate displays. Initialise total before the **loop** so a zero-**iteration** run still has a defined value.

Limit: This finite **loop** does not establish behavior of a while-**loop** with a different update or stopping condition.

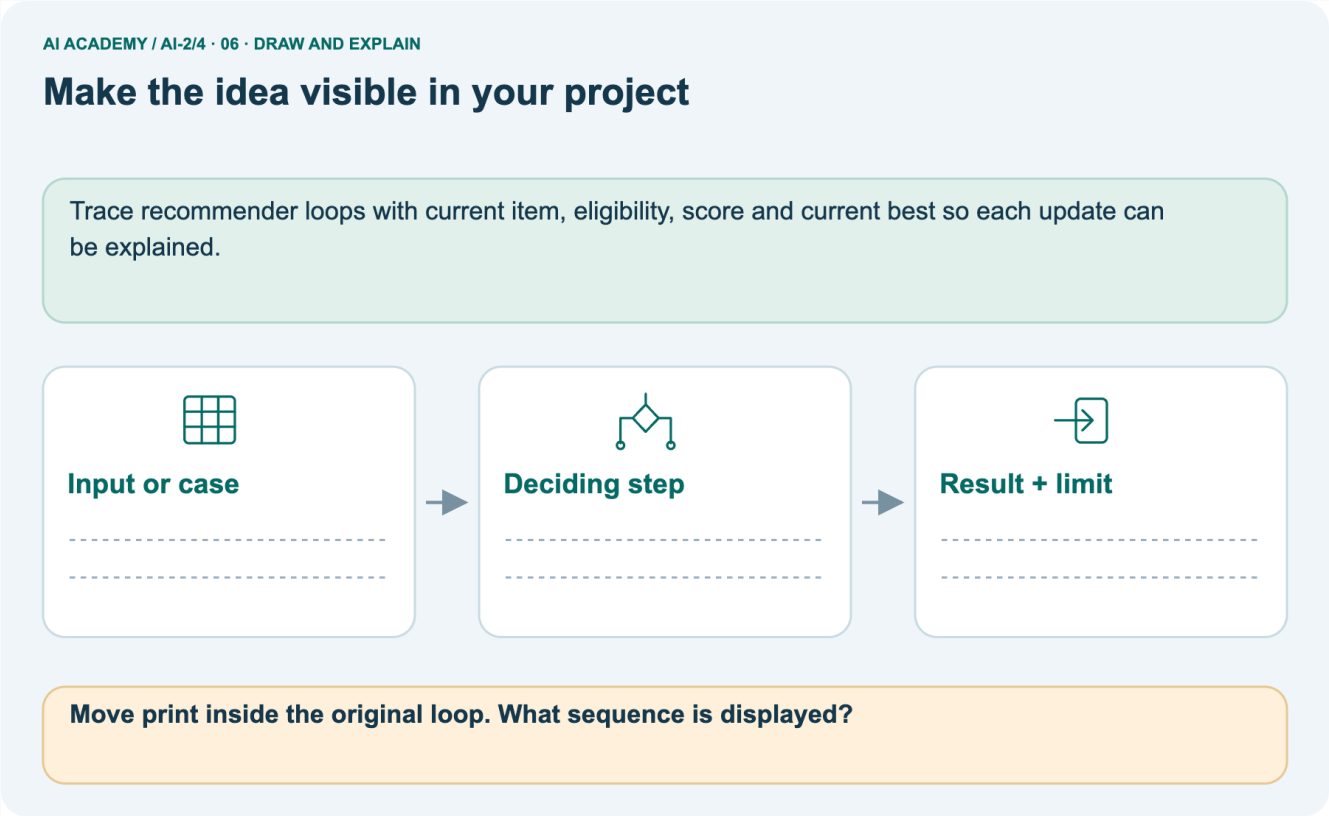
Explain the case, then point to a limit on the claim.

Discuss while showing the picture

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the **state**d scope: This finite **loop** does not establish behavior of a while-**loop** with a different update or stopping condition.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Trace recommender **loop**s with current item, eligibility, score and current best so each update can be explained. Draw your **input**, deciding step, result and limit. Supported question: Move print inside the original **loop**.

What sequence is displayed?

Draw your project **trace** and justify the deciding step using **evidence**.

Discuss while showing the picture

Teacher prompt: Draw your project **trace** and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: 4,7,10,13 on separate lines, because each print follows one update.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.